

Tyler M. Chinsky

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EDUCATION

M.S., Bioinformatics, Johns Hopkins University | Baltimore, MD

May 2025

B.S., Bioinformatics and Computational Biology, University at Buffalo | Buffalo, NY

Feb 2022

Cum Laude

TECHNICAL SKILLS

Genomic & Bioinformatic Analysis: Next-generation sequencing (NGS), bulk/single-cell/spatial RNA-seq, multiome, ChIP-seq/ATAC-seq, population and individual variant calling (short & structural variants), genome/transcriptome-wide association analysis (GWAS/TWAS), OncoPanel analysis, GSEA/GSVA, clinical biomarker discovery, immunogenomics, immunoncology, structural & functional protein analysis

Programming & Pipelines: Python, R, WDL, Nextflow, Docker/Apptainer (Docker & Kubernetes certified), Terra, Google Cloud Platform, cloud computing, high-performance computing (HPC), machine learning (ML), agentic AI, C++, Java, Scala

Wet Lab: Next-generation sequencing (NGS) library prep (Illumina NextSeq/NovaSeq), DNA/RNA extraction, PCR, protein purification, multiplex immunofluorescence imaging

RESEARCH EXPERIENCE

Computational Biologist | Genomics and Bioinformatics Core, University at Buffalo

2024 - Present

- Lead analysis projects for internal and external customers spanning spatial biology, single-cell, bulk, and multiome sequencing
- Lead development and integration of scalable, containerized bioinformatics workflows (Docker/Apptainer/Nextflow) for standard core analyses
- Lead integration of agentic AI and ML tooling into core workflows to improve analyst efficiency and reduce project turnaround time
- Contributed to multiple peer-reviewed publications arising from core-supported research projects

Computational Biologist (Consultant) | D2Dx Consulting

2024

- Analyzed clinical and molecular data to identify candidate biomarkers supporting oncology clinical trial design
- Applied GSVA/GSEA methods to evaluate gene expression signatures against clinical and molecular annotations

Computational Biologist | Dept. of Breast Oncology, Dana-Farber Cancer Institute

2022 - 2024

Advisor(s): Saud AlDubayan, Eliezer Van Allen

- Independently led multiple projects in cancer genomics, including characterizing the somatic genetic architecture of ER+ metastatic breast cancer, evaluating the influence of germline variation on disease presentation, and associating tumor genetic characteristics with immunotherapy response
- Performed large-scale genomic analysis using cloud computing platforms (Terra, Google Cloud) and developed more than 10 documented workflows for sequence alignment, population and individual variant calling (short and structural variants), OncoPanel analysis, genome- and transcriptome-wide association analysis, and statistical analysis, shared across the team
- Built and maintained data processing pipelines for high-throughput clinical DNA and RNA sequencing, with an emphasis on immune profiling
- Extracted and analyzed spatial data from multiplex immunofluorescence imaging in support of collaborative, co-authored oncology studies
- Expanded statistical power of ongoing projects by sourcing data through collaborator outreach and curation of public repositories (e.g., Sequence Read Archive); presented research progress at lab meetings and conferences, and assisted with onboarding new team members

Affiliate Scientist | Cancer Program, Broad Institute of MIT and Harvard

2022 - 2023

- Leveraged Broad Institute computational resources and datasets to support oncology research conducted at Dana-Farber Cancer Institute

Computational Biologist | SUNY CTRC, University at Buffalo

2020 - 2023

Advisor(s): Animesh Sinha, Kristina Seiffert

Pemphigus vulgaris (PV) HLA binding site identification and genomic expression analysis using NGS

- Identified HLA alleles as disease drivers of pemphigus vulgaris using predisposition markers, microarray sequencing, and functional analyses; discovered gene sets that categorize by disease and/or HLA status, including protective signatures, and pathway mutations associated with elevated polygenic risk score
- Built and executed pipelines in Python, R, and Bash to analyze microarray, RNA-seq, scRNA-seq, and genomic data, filtering noise from raw reads and comparing expression profiles across disease and control cohorts
- Designed experiments to functionally validate NGS-derived signatures, focusing on transcriptome profiling of immunological pathways
- Contributed to collaborative, co-authored publications

Associate Research Technician | Yu Laboratory, University at Buffalo

2021 - 2022

Advisor(s): Michael Yu

- Investigated arginine methylation in RNA of budding flocculent yeast in the presence of chromatin remodeling proteins
- Conducted in vivo experiments verifying HMT protein methylation levels in the presence and absence of Snf2, Ime4, and eIF1A
- Applied ex vivo techniques to analyze protein expression of HMT proteins across genetic backgrounds

Associate Computational Biologist | Koudelka Laboratory, University at Buffalo

2021

Advisor(s): Gerald Koudelka

- Identified consensus regions within Shiga toxin-producing lambda bacteriophage DNA
- Constructed R programs to support secondary structure prediction analysis

Associate Computational Biologist | Cui Laboratory, Rochester Institute of Technology

2015 - 2016

Advisor(s): Feng Cui

Chromatin organization associated with DNA binding of tumor suppressor TP53

- Developed ChIP-seq analysis pipelines to identify genomic regions inhibiting TP53 DNA-binding sites
- Analyzed gene expression data for pathway activation/inhibition at specific TP53 binding regions

Laboratory Technician Assistant | Rochester Institute of Technology

2018 - 2019

- Prepared, set up, and maintained instructional laboratories, including media, reagents, and buffers, in accordance with departmental and regulatory standards
- Maintained bacterial and eukaryotic cell cultures for instructional use

TEACHING EXPERIENCE

Associate Research Mentor | Yu Laboratory, University at Buffalo

2021 - 2022

- Mentored two undergraduate students in laboratory technique and experimental design alongside the principal investigator
- Trained mentees on scientific notebook-keeping practices, including digital recordkeeping of figures, images, and tables

Instructor, Computer Security, Data Analysis & Machine Learning | iD Tech Camps

2018

- Designed and taught coursework introducing high school students to Python, C++, and Java
- Taught approximately 20 students per week over a 12-week summer program using varied instructional approaches

PUBLICATIONS

Peer-Reviewed Publications

Leick, M.B., Han, S., Gallagher, K.M.E., ... **Chinsky, T.M.**, ... Van Allen, E., AlDubayan, S., Maus, M.V. (2026). Genomic correlates of clinical CAR T cell activity. *Science Immunology*, 11(121), eaef4134.

Bowitch, A., **Chinsky, T.M.**, Yu, M.C., Ferkey, D.M. (2022). The C. elegans OCTR-1 and human alpha-2A adrenergic receptors are methylated within the third intracellular loop by human PRMT5 in vitro. *microPublication Biology*. doi:10.17912/micropub.biology.000546

Parthasarathy, A., Wong, N.H., Weiss, A.N., ... **Chinsky, T.M.**, ... Hudson, A.O. (2019). SELfies and CELLfies: whole genome sequencing and annotation of five antibiotic resistant bacteria isolated from the surfaces of smartphones, an inquiry-based laboratory exercise in a genomics undergraduate course at the Rochester Institute of Technology. *Journal of Genomics*, 7, 26-30.

Peer-Reviewed Abstracts & Presentations

Hamid, A.A., Vergara, M., Park, J., **Chinsky, T.M.**, ... Sweeney, C.J. (2024). Whole exome correlates in metastatic hormone sensitive prostate cancer (mHSPC): results from E3805 CHARTED. *J Clin Oncol*, 42, 5088.

Hamid, A.A., **Chinsky, T.M.**, Vergara, M., Grist, E., Attard, G., Van Allen, E., AlDubayan, S., Sweeney, C.J. (2023). Landscape and impact of germline pathogenic variants (PVs) in metastatic hormone sensitive prostate cancer (mHSPC): ancillary study of E3805 CHARTED. *J Clin Oncol*, 41(16_suppl), 5082.

Chinsky, T.M., Chen, R., Holan, S., Seiffert, K.R., Sinha, A.A. (2022). Investigation of transcriptionally regulated differential gene expression as a potential protection signature in the pemphigus vulgaris disease model. SUNY Buffalo Jacobs School of Medicine Research Symposium [Poster/Abstract].

CERTIFICATIONS

Docker Mastery with Swarm and Kubernetes Udemy	Jan 2024
AWS Knowledge: Cloud Essentials Amazon Web Services (AWS)	Jun 2025
Docker Swarm Mastery: DevOps Style Cluster Orchestration Udemy	Jun 2026

PROFESSIONAL MEMBERSHIPS

American Association for Cancer Research (AACR)	2024 – Present
International Society for Computational Biology (ISCB)	2023 – Present

HONORS AND AWARDS

Latin Honors - Cum Laude, University at Buffalo SUNY, 2022
Dean's List, SUNY Empire State, 2020
Emerson Student Undergraduate Research Fellowship, Rochester Institute of Technology, 2016
FIRST Robotics Scholarship, Rochester Institute of Technology, 2015-2018